

EXPERIMENT 15

DCL Commands – GRANT, REVOKE

Aim:

To learn how to use various DCL commands such as GRANT and REVOKE.

Procedure and Syntax:

Data Control Language (DCL) is used to control privileges in Database. To perform any operation in the database, such as for creating tables, sequences or views, a user needs privileges. Privileges are of two types, **System:** This includes permissions for creating session, table, etc and all types of other system privileges.

Object: This includes permissions for any command or query to perform any operation on the database tables.

In DCL we have two commands,

GRANT: Used to provide any user access privileges or other privileges for the database.

REVOKE: Used to take back permissions from any user.

- Allow a User to create session
- When we create a user in SQL, it is not even allowed to login and create a session until and unless proper permissions/privileges are granted to the user.
- Following command can be used to grant the session creating privileges.

Questions

1. Create a new MySQL user and grant the user permission to **SELECT** data from the Patient table.
2. Grant the user permission to **INSERT** records into the Patient table.
3. Revoke the **INSERT** permission from the user.

OUTPUT:

1)

```
1 • USE Hospital1;
2
3 • CREATE USER IF NOT EXISTS 'patient_user'@'localhost'
4 IDENTIFIED BY 'Patient@123';
5
6 • GRANT SELECT
7 ON Hospital1.Patient
8 TO 'patient_user'@'localhost';
9
10 • SHOW GRANTS FOR 'patient_user'@'localhost';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Grants for patient_user@localhost
▶	GRANT USAGE ON *.* TO `patient_use...
	GRANT SELECT ON `hospital`.`patient...

2)

```
11 • USE Hospital1;
12
13 • GRANT INSERT
14 ON Hospital1.Patient
15 TO 'patient_user'@'localhost';
16
17 • SHOW GRANTS FOR 'patient_user'@'localhost';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Grants for patient_user@localhost
▶	GRANT USAGE ON *.* TO `patient_use...
	GRANT SELECT, INSERT ON `hospital`....

3)

```
18 • USE Hospital1;  
19  
20 • REVOKE INSERT  
21 ON Hospital1.Patient  
22 FROM 'patient_user'@'localhost';  
23  
24 • SHOW GRANTS FOR 'patient_user'@'localhost';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Grants for patient_user@localhost	
▶	GRANT USAGE ON *.* TO `patient_use...
	GRANT SELECT ON `hospital1`.`patient...

RESULT:

Thus the DCL commands GRANT and REVOKE SQL executed successfully.